# Seydina Ousmane **NIANG**

Phd Student

## About me

I am a PhD student in applied mathematics, specializing in **graphs theory**. My research focuses on **deep probabilistic graphical models** for the analysis of **networks** with **continuous attributes**.

## Contact


Born on 08/09/1998

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Nice , France

Seydina Ousmane NIANG


## Languages

Wolof - Native Language

French - Fluent

English - Fluent

## Soft Skills and Strengths

Creativity | Curiosity | Flexibility | Self Confidence | Ability to Plan and Organize | Autonomy | Team Working | Leadership | Good Listener | Patience

## Professional Skills

Pytorch | Tensorflow | Keras | Scikit-Learn | Python/R

## Other Interests

- Taekwondo
- Gym
- Football
- Movies

## EDUCATION

**Nov-2023-Ongoing**

**Phd in applied mathematics** — Nice , France

**Côte d'Azur University**

Deep latent variable models for the joint analysis of networks and continuous data

**2022-2023**

**M2 Data Science - Artificial Intelligence** — Paris , France

**Ecole Polytechnique Paris-Saclay University**

*Data Science*

deep learning | reinforcement learning | statistical learning | optimization | big data | graph theory | NLP

**2021-2022**

**M1** — Paris France

**Paris- Saclay University**

*Applied Mathematics*

statistics | optimization | machine learning | probabilities | time Series | operational research | deep learning

**2020-2021**

**Magistere Degree** — Paris France

**Paris-Saclay University**

*Pures Mathematics*

Strong knowledge of fundamental mathematics including probability, statistics, dynamical systems and algebra

**2020-2021**

**Bachelor** — Paris France

**Paris-Saclay University**

*Double bachelor mathematics and informatics*

Basics on mathematics and computer sciences

## WORK EXPERIENCE

**May/Sep-2022**

**Internship Deep Learning:Neural Differentials Equations** — Paris, France

*Laboratoire de Mathématiques d'Orsay*

Neural Differential equation | Pharmacokinetics | Pytorch | torchdyn | Pytorch Lightning

**February/June-2020**

**Supervised Research Work** — Paris, France

*Notion of free groups on any set X and properties of their subgroups*

**Nov-2023-Ongoing**

**PhD** — Nice, France

*Deep generative models for the joint analysis of network and continious data*

INRIA, MASSAI

## Publications

**NIANG Seydina Ousmane et al. (Jan 2025)** "The Deep Zero-Inflated Latent Position BlockModel for the Clustering of Nodes in Graphs"

**NIANG Seydina Ousmane et al. (May 2025)** "Importance weithing graph variational autoencoder for the cliustering of nodes in complexe networks "

**NIANG Seydina Ousmane et al. (Nov 2024)** "The Deep Latent Position Block Model for the Clustering of Nodes in Multi-Graphs"

**NIANG Seydina Ousmane et al. (May 2024)**"Conditional Denoising Diffusion Probabilistic Models for the Clustering of Images"

**NIANG Seydina Ousmane et al. (Dec 2025)**"Temporal variational graphon autoencoder for rupture detecion "